Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/27/2006
PI Name:	Rowe, David W M.D.		
Project Title:	Use of GTP Transgenes to Assess Osteoprogenitor in Intact Bone		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Farmington	**State:**	CT
Zip Code:	06032	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-02
Start Date:	09/01/2000	**End Date:**	08/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG2-1434		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The uniting objective of this grant is to demonstrate that a promoter Green Fluorescent Protein (GFP) transgene that activates at a specific stage in the osteoblastic lineage can be developed into a sensitive tool for assessing the cellular response of bone as integrated organ to mechanical loading. Because we are a molecular biology and not histology laboratory, there were a number of technical hurdles that we had to master before the validity of our concept could be demonstrated. Thus the grant was divided into a first aim that was technology and methods driven and a second and third aim that applied this methological skill to two models of mechanical loading, orbital shaking and tail left. The first year the grant was used to acquire the new technologies and to fabricate the equipment necessary to apply mechanical loading to mice. The second-year has been used to better understand the technologies that we have acquired and to apply them to our two model systems.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Bone adapts to its mechanical environment by altering the tempo and magnitude of the bone remodeling dynamics that exists between the osteoprogenitor and osteoclast lineage. Mice can be engineered to bring out the fine details of this interrelationship and experiments can be performed to appreciate the microheterogeneity of the two lineages within intact bone in a quantitative manner. We are fortunate to have been given the resources to develop the mice and the microscopic techniques to appreciate these changes and believe that we have the appropriate technology and models for this analysis. There is learning curve for us to develop facility in the use of these techniques and models. We hope that in the 02 year the grant that many of these new initiatives will converge to validate the utility of our proposed approach.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)